Math 571 - Homework 4

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Problem 4.1 (B:4.2:2)**.** Show that the following are equivalent for $f : X \to Y$, where X and Y are metric spaces.

(a) f is continuous.

(b) If (p_n) is a sequence from X and $p_n \to p$, then $f(p_n) \to f(p)$, equivalently, $\lim_n f(p_n) = f(\lim_n p_n)$.

(c) If (p_n) is convergent, then $f(p_n)$ is convergent.

Hint: For (c) $\implies$ (b) let $q = \lim_n f(p_n)$ and consider $(p_0, p, p_1, p, p_3, p, \ldots) = (\hat{p}_n)$, then $(f(\hat{p}_n)) = (f(p_0), f(p), f(p_1), f(p), \ldots)$ must converge.

Problem 4.2 (B:4.2:3)**.** Let X be a metric space.

(a) Show that there are open subsets A, B of X, such that $A \cup B = X$ and $A \cap B = \emptyset$ if and only if there exists a continuous function $f : X \to R$ such that $f(a) = 0$ for all $a \in A$ and $f(b) = 1$ for all $b \in B$.

(b) Deduce that X is connected if and only if every continuous function $X \to \{0, 1\}$ is constant.

(c) Show likewise that X is connected if and only if every continuous function $X \to \mathbb{Z}$ is constant.

Problem 4.3 (B:4.2:4)**.** (a) Suppose $f : \mathbb{R} \to \mathbb{R}$ is a continuous function such that $f(x) > x$ for all x. For any positive integer n, let f^n denote the n-fold composite $f \circ f \circ \ldots \circ f$. Show that for all $x, y \in R$, there exists a positive integer n such that $f^n(x) > y$.

(b) Does the conclusion of part (a) remain true if the assumption that f is continuous is removed?

Problem 4.4 (R:4.11)**.** Suppose f is a uniformly continuous mapping of a metric space X into a metric space Y and prove that $(f(x_n))$ is a Cauchy sequence in Y for every Cauchy sequence (x_n) in X. Use this result to give an alternative proof of the theorem stated in Exercise 13.

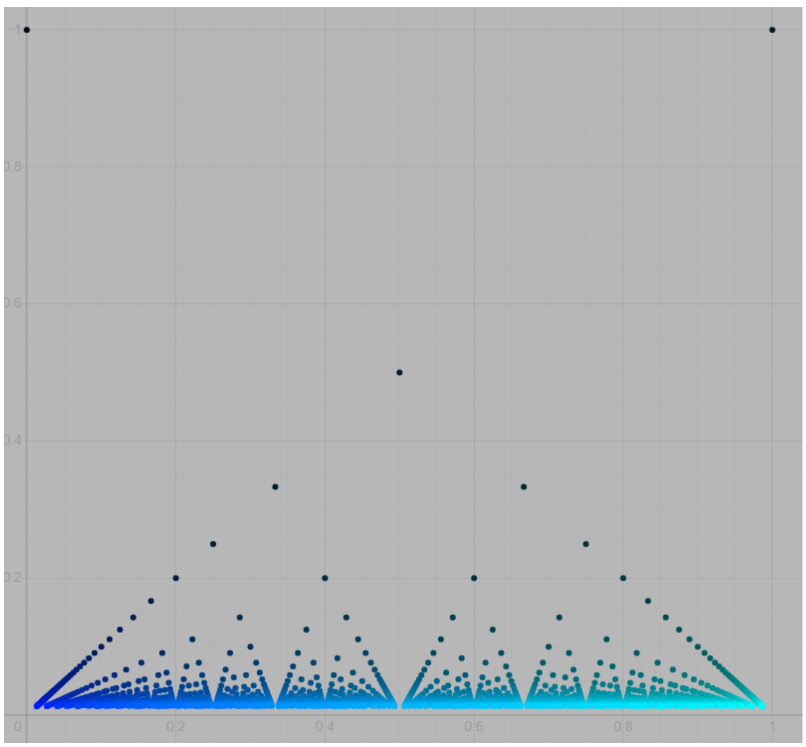
Problem 4.5 (R:4.13)**.** Let E be a dense subset of a metric space X, and let f be a uniformly continuous *real* function defined on E. Prove that f has a continuous extension from E to $\text{cl}(E)$. (This means, there is a unique continuous $\hat{f} : \text{cl}(E) \to Y$ so that $\hat{f} \supset f$.)

Could the range space $\mathbb{R}^1$ be replaced by $\mathbb{R}^k$? By any compact metric space? By any complete metric space? By any metric space?

Better Hint: Use R:4.11

Problem 4.6 (R:4.18)**.** Define the ***Thomae Function*** as follows

$$f(x) = \begin{cases} 0 & x \text{ is irrational} \\ \frac{1}{n} & x = \frac{m}{n}, \text{ where } n, m \in \mathbb{Z}, n > 0, \text{ and } \gcd(m,n) = 1 \end{cases}$$



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(a) Show that f is discontinuous at exactly the rational points.

(b) Show that for $g : \mathbb{R} \to \mathbb{R}$, $\text{Dis}(f) = \{x \mid f \text{ is discontinuous at } x\}$ is an F_σ set. (A union of countably many closed sets.)

Hint: Look at B:4.3:7 and the ***oscillation*** of g at x, denoted $\omega_f(x)$. Show that $\text{Dis}(f) = \bigcup D_n$ where $D_n = \{x \mid \omega_g(x) \geq \frac{1}{n}\}$ and that D_n is closed.

(c) Show that there is no $g : \mathbb{R} \to \mathbb{R}$ that is continuous at exactly the irrational points. (Baire Category)

Problem 4.7 (B:4.4:3)**.** Amazingly, there are continuous functions $f : [0,1] \to [0,1]^k$ for any k. Here are a couple of nice videos on this topic:

- A great 3Blue1Brown video: *Hilbert's Curve: Is infinite math useful?*
- Eddie Woo discussing several constructions in a high school class: *Space-Filling Curves (1 of 4: Peano Curve)*

(a) Let $n > 1$. Show that for every point p of $[0,1]^n$, the subset obtained by removing p from $[0,1]^n$ is still connected. On the other hand, point to a result in Rudin showing that $[0,1]$ does not have this property.

(b) Show how the existence of a one-to-one continuous map from $[0,1]$ onto $[0,1]^n$, together with the above pair of facts, would lead to a contradiction. (Hint: What do you know about one-to-one continuous maps of one compact metric space onto another?)

Problem 4.8 (B:4.3:4)**.** (a) Suppose K is a compact metric space, and $f : K \to K$ is a map with the property that for every pair of distinct points $x, y \in K$ one has $d(f(x), f(y)) < d(x,y)$. Show that there exists a unique point $p \in K$ such that $f(p) = p$.

(b) Show by examples that the result of (a) is false if we put any of the noncompact metric spaces $[0,1)$, $[0,+\infty)$ or $\mathbb{R}$ in place of the compact space K.

Problem 4.9 (R4.20/22)**.** Let X be a metric space, and for any nonempty subset $E \subset X$, define the distance from a point $x \in X$ to E as $\rho_E(x) = \inf_{z \in E} d(x,z)$.

(a) Prove that $\rho_E(x)$ is a uniformly continuous function on X, and that $\rho_E(x) = 0$ if and only if $x \in \overline{E}$.

(b) Let A and B be disjoint, nonempty closed sets in X. Define:

$$f(p) = \frac{\rho_A(p)}{\rho_A(p) + \rho_B(p)}$$

Prove that f is a continuous function on X with range in $[0,1]$, such that $f(p) = 0$ precisely on A and $f(p) = 1$ precisely on B.

You could skip this, but it adds context.

The next problem uses a tool that is also at the heart of some of the space-filling curves and is a good capstone to the techniques in this chapter. Consider the space $C([0,1]) = \{f : [0,1] \to \mathbb{R} \mid f \text{ is continuous}\}$. (You have studied this as an inner product space in MAT-512). X is provided with a norm, the L_∞***-norm*** or ***uniform*** norm, which is different from the L_2-norm, $\|f\|_2 = \left(\int_0^1 f^2\,dx\right)^{1/2}$ coming from the inner-product $\langle f, g\rangle = \int_0^1 fg\,dx$. The uniform norm is given by

$$\|f\|_1 = \sup\{|f(x)| \mid x \in [0,1]\}$$

This gives rise, as all norms do, to a metric, $d_1 : C([0,1]) \times C([0,1]) \to [0,\infty)$ given by $d_1(f,g) = \|f-g\|_1 = \sup\{|f(x)-g(x)| \mid x \in [0,1]\}$. It should be clear that the 1/2-norms and the resulting metrics are just the natural generalization to $C([0,1]) \subset \mathbb{R}^{[0,1]}$ of the corresponding norms/metrics on $\mathbb{R}^k$. (In fact, this is what you naturally do when approximating continuous functions by taking test points, a technique common to both Calculus and computer science.)

Note: By the extreme value theorem, $\sup\{|f(x)| \mid x \in [0,1]\}$ can be replaced by $\max\{|f(x)| \mid x \in [0,1]\}$.

Convergence in $C([0,1])$ under the L_∞-norm is called ***uniform convergence***, i.e., convergence in the uniform norm. Clearly, $f_i \to f$ in $C([0,1])$ iff for all $\epsilon > 0$, there is $N > 0$, so that for all $i > N$, $\sup\{|f_i(x) - f(x)| \mid x \in [0,1]\} < \epsilon$.

Fact: $C([0,1])$ under the 1-norm metric is complete.

Proof: It is actually an excellent exercise to prove this. Suppose (f_i) is a Cauchy sequence. Then $(f_i(x))$ is a Cauchy sequence for each x ans since $\mathbb{R}$ is complete, $f_i(x) \to y$ for each x, so consider $f(x) = \lim_i f_i(x)$. To complete the argument, one only needs to show that $f_i \to f$ in the 1-norm, i.e., uniformly. Fix $\epsilon > 0$ and N so that $\|f_i - f_m\|_1 < \epsilon/2$ for $i, m > N$. Then $|f_i(x) - f(x)| \le |f_i(x) - f_m(x)| + |f_m(x) - f(x)|$. By choosing $m > N$ with $|f_m(x) - f(x)| < \epsilon/2$ we see that $|f_i(x) - f(x)| < \epsilon$ for all $i > N$ for all $x \in [0,1]$. But then $\|f_i - f\|_1 < \epsilon$ for all $i > N$ so $f_i \to_1 f$.

You should stop skipping if you chose to skip above.

We can rephrase uniform convergence as $f_i \to f$ uniformly if

$$(\forall \epsilon > 0)(\exists N > 0)(\forall i > N)(\forall x \in [0,1])(|f_i(x) - f(x)| < \epsilon) \quad (\dagger)$$

Note that $f_i(x) \to f(x)$ ***pointwise*** iff

$$(\forall \epsilon > 0)(\forall x \in [0,1])(\exists N > 0)(\forall i > N)(|f_i(x) - f(x)| < \epsilon)$$

So uniform convergence means we can find for each ϵ and N that ***works uniformly*** for all $x \in [0,1]$.

Main Result: If f_i are continuous and $f_i \to f$ uniformly in the sense of ($\dagger$), then f is continuous.

Proof. This would have made a nice exercise in itself. For $\epsilon > 0$ and take $N > 0$ so that for all $i > N$, $|f_i(x) - f(x)| < \frac{\epsilon}{3}$ for all $x \in [0,1]$. Take $i > N$ and find δ so that $0 < |y - x| < \delta \implies |f_i(y) - f_i(x)| < \frac{\epsilon}{3}$. Now take $0 < |y - x| < \delta$, then

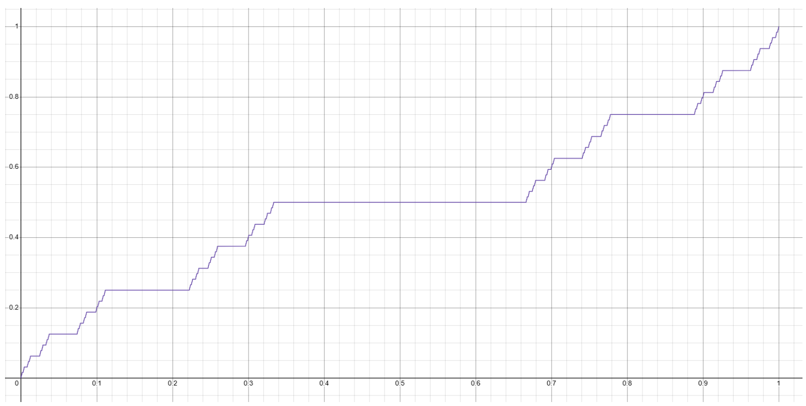
$$|f(y) - f(x)| \le |f(y) - f_i(y)| + |f_i(y) - f_i(x)| + |f_i(x) - f(x)| < \epsilon$$

by the triangle inequality. So we have verified that

$$0 < |y - x| \implies |f(y) - f(x)| < \epsilon$$

Note: If $f_i \to f$ pointwise, but not uniformly, then f need not be continuous. Consider $f_n(x) = x^n$ on $[0,1]$, $f(x) = 0$ for $x \in [0,1)$, but $f(1) = 1$, so f is not continuous. This shows that $f_n \to f$ pointwise, but not uniformly.

Problem 4.10 (**The Cantor Function**)**.** Let $C \subseteq [0,1]$ be the standard Cantor ternary set. Let $f_n : [0,1] \to [0,1]$ be the sequence of continuous, monotonically increasing functions where $f_0(x) = x$, and each f_{n+1} is constructed by modifying f_n so that it is constant on the middle thirds removed in the $(n+1)$-th step of the Cantor set construction.



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(a) Prove that the sequence $\{f_n\}$ converges uniformly on $[0,1]$ to a limit function f (the Cantor function).

  Hint: You may use that $C([0,1])$ is complete as a metric space using the uniform metric. This way it suffices to show $\|f_n - f_m\|_\infty < \epsilon$ for sufficiently large m, n.

(b) Prove that f is continuous and monotonically non-decreasing on $[0,1]$.

(c) Prove that f is surjective onto $[0,1]$, yet is locally constant on the complement of C (an open set of measure 1).